

## GLOBAL INDUSTRY CHALLENGE 2026

*Dynamic Systems Forecasting*

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### Quantum Reservoir Computing for Financial Regime Intelligence

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*Track: Dynamic Systems Forecasting: Financial Volatility Prediction*

Team: **QuantumEdge**

| Name                | Role                                     | Background & Affiliation                                                              |
|---------------------|------------------------------------------|---------------------------------------------------------------------------------------|
| Judith Ann Sarjeant | Coder, Finance Lead, PM                  | Doctoral Student; Qiskit, Python, IBM Quantum, Quantum ML, Finance                    |
| Shema Nourine       | Technical Lead                           | M.Sc Applied Mathematics; Qiskit, Python                                              |
| Hasarindu Perera    | Coder, Commercial Lead                   | BSc Management and IT; Software Engineer (AI); Qiskit, PennyLane, PyTorch             |
| Van Tien Nguyen     | Team Lead - Coder, Data / Technique Lead | PhD Student in Quantum ML; Qiskit, PennyLane, PyTorch; VQA, QNN design & benchmarking |

## 1. Team Qualifications

QuantumEdge covers all key GIC roles. **Sarjeant** (Finance Lead & Project Manager): DBA Student, Digital Finance, Quantum ML, Qiskit, Python, IBM Quantum. **Nourine** (Technical Lead): MSc in Applied Mathematics; thesis on quantum fidelity metrics; Qiskit experience. **Perera** (Coder/Domain Expert): ML researcher in RL, multi-agent systems, and QML; leads classical baselines. **Nguyen** (Coder/Research Lead): PhD student in Quantum ML; experience in PennyLane, PyTorch, VQA, QNNs, benchmarking, and hybrid quantum-classical experimentation. All members have hands-on experience with qBraid, Qiskit, and PennyLane on IBM hardware.

## 2. Solution Steps

(1) Collect S&P 500 OHLCV, CBOE VIX, and Oxford-Man realized-volatility data; (2) engineer features (returns, realized volatility, Garman–Klass estimator, VIX–RV spread) and label regimes via Gaussian HMM; (3) encode inputs with RY-angle rotations into two fixed-Hamiltonian quantum reservoirs; (4) extract Pauli-expectation vectors augmented with entanglement-spectrum eigenvalues; (5) train a ridge-regression readout on concatenated feature vectors; (6) benchmark against ARIMA, LSTM, Echo State Network, and XGBoost; (7) validate reservoir expressivity with the required MNIST benchmark across 5, 10, and 15 qubits. The architectural pipeline facilitates a systematic transition from raw market signals to regime-aware intelligence, ensuring end-to-end reproducibility across all computational stages. This synthesis of financial feature engineering, quantum feature extraction, and rigorous classical benchmarking ensures that both predictive performance and the specific quantum contribution are assessed with methodological clarity.

## 3. High-Level Overview

Financial volatility operates on multiple timescales—fast shocks and slow regime clustering. Classical HAR models capture this via temporal lag decomposition. We bring HAR-style decomposition into the quantum domain: a **dual-reservoir QRC** where a short-term reservoir (weak Ising coupling, shallow depth) captures rapid shocks and a long-term reservoir (stronger coupling, oscillatory components) tracks persistent regime momentum. Their feature vectors are fused at a single linear readout, achieving regime detection without any gradient-based quantum training. This architecture maintains the inherent simplicity and noise-robustness of reservoir computing while exploiting quantum dynamics to generate *high-dimensional temporal embeddings*. Consequently, the model is uniquely calibrated for financial environments requiring rapid adaptation, nonlinear representation, and negligible training overhead.

## 4. Technical Approach & Quantum Advantage

**Architecture.** Two parallel fixed-Hamiltonian (Ising-type) reservoirs use RY-angle encoding—a no-gradient design that eliminates barren plateaus. The readout combines Pauli expectations with *entanglement-spectrum features* (eigenvalues of each reservoir’s reduced density matrix), which encode multi-body quantum correlations provably inaccessible to classical reservoirs—the core quantum advantage claim. Only the ridge-regression readout is trained; no quantum parameters are differentiated.

**Feasibility.** NISQ-compatible at 5–15 qubits. Noise robustness validated with depolarizing and amplitude-damping channels calibrated to IBM Eagle r3 / Heron r1. MNIST benchmark (Fujii et al.) validates expressivity across qubit counts for cross-team comparison. Classical baselines match Li et al. (2025) for comparability. A qBraid Skill will be submitted alongside the GitHub repository for reproducibility. Metrics: RMSE, regime-classification accuracy, and Sharpe-ratio improvement.

## 5. Projected Industry Impact

**Portfolio risk management:** dual-timescale signals provide earlier drawdown warning, enabling pre-emptive position sizing and cash-buffer increases. **Derivatives pricing:** regime-aware vol forecasts support real-time Heston/SABR parameter adjustment around earnings and macro events, improving mark-to-market accuracy. **Algorithmic execution (JonesTrading):** regime meta-signals shift strategies from passive (TWAP/limit orders) to aggressive (immediate fills) precisely when market conditions warrant speed, directly reducing implementation shortfall for institutional clients.